Technical Report: Phase 1 & Phase 2 Detailed Implementation

Project: Machine Learning-Based Evapotranspiration (ET0) Estimation for Irrigation Support | **Region:** Hisar, Haryana

5,113 Days	14 Years	0 Missing Data	FAO-56 Verified
Daily Observations	2010 - 2023 Span	100% Quality Audit	pyet Agreement (<0.00012 mm)

1. Project Overview & Problem Statement

Evapotranspiration (ET0) is the combined process of soil water evaporation and crop plant transpiration. It is the international benchmark for determining crop water requirements and setting irrigation pumping schedules. In intensive agricultural regions such as Northern India (Haryana and Punjab), groundwater depletion is acute, making accurate daily ET0 estimation essential for precision water management.

The standard benchmark for estimating ET0 is the **FAO-56 Penman-Monteith equation** (Allen et al., 1998). However, this equation requires full meteorological records: maximum and minimum temperatures, relative humidity, wind speed, solar radiation, and atmospheric pressure. In rural agricultural weather stations, **precision pyranometers (solar radiation sensors, costing $1,000 - $2,500 USD)** and anemometers are frequently unavailable, uncalibrated, or broken due to dust and weather exposure.

Core Research Hypothesis: Can machine learning regression models (Linear Regression, Random Forest, SVR) accurately approximate FAO-56 ET0 using only basic, low-cost sensors (temperature and relative humidity), thereby reducing hardware costs by over 90% without compromising irrigation accuracy?

2. Phase 1: Data Acquisition, Cleaning & Quality Auditing

Study Area: The station coordinates correspond to **Hisar, Haryana, India (Latitude: 29.15 N, Longitude: 75.72 E, Elevation: 215 meters)**. This site was selected because Hisar is the agricultural research capital of Haryana (home to CCS Haryana Agricultural University) and directly matches the regional scope of the foundational study (*Shaloo et al., Journal of Atmospheric and Solar-Terrestrial Physics*).

Data Source & Extraction: Daily agroclimatology data spanning 14 full calendar years (**January 1, 2010 to December 31, 2023**) was obtained from the **NASA POWER API** (Prediction of Worldwide Energy Resources), which combines GEOS/MERRA-2 atmospheric data and CERES satellite solar irradiance.

Step-by-Step Data Engineering Pipeline:

- **Automated Header Stripping:** NASA POWER CSV files contain a 14-line metadata header (-BEGIN HEADER- to -END HEADER-). Our loader scans line-by-line, dynamically locates the header termination line, and loads the pure tabular rows starting from line 15.
- **Date Indexing:** Combined NASA's YEAR and DOY (Day of Year 1-365) columns using Python's pd.to_datetime(format='%Y%j') to generate real calendar dates (YYYY-MM-DD) set as the chronological index.
- **Missing Value Audit:** Audited all records for NASA fill codes (-999.0). The Hisar dataset contained **0 missing records (100% data completeness across all 5,113 days)**.
- **Physical Consistency Bounds:** Validated that Max Temperature >= Min Temperature, Relative Humidity remained within [0%, 100%], and Wind Speed remained non-negative (>= 0 m/s).
- **Cross-Verification:** Verified the programmatically fetched dataset against a manual browser download, confirming a **100.0% row-by-row exact match**.

Table 1: Meteorological Variables & Statistical Ranges (5,113 Daily Observations)

Variable Name	Dataset Column	Unit	Min	Mean +/- Std	Max	Physical Meaning in Agriculture
Max Temperature	max_temp	deg C	13.44	33.15 +/- 7.20	47.89	Peak daytime air temperature; drives thermal vapor deficit
Min Temperature	min_temp	deg C	0.25	19.49 +/- 8.18	34.69	Nighttime minimum; approximates daily dewpoint temperature
Mean Temperature	mean_temp	deg C	6.67	25.87 +/- 7.92	40.90	Average of daily high and low; controls saturation curve slope
Relative Humidity	relative_humidity	%	7.72	42.29 +/- 17.76	90.17	Moisture saturation; drops <10% in summer, >90% in winter fog
Wind Speed (2m)	wind_speed	m/s	0.45	1.89 +/- 0.79	6.08	Aerodynamic transport of humid boundary layer away from crops
Solar Radiation	solar_radiation	MJ/m2/d	1.40	17.52 +/- 5.69	29.92	Incoming shortwave energy that drives water evaporation
Surface Pressure	surface_pressure	kPa	96.67	98.30 +/- 0.70	99.83	Barometric pressure for psychrometric constant calculation
Target ET0	**ET0**	**mm/day**	**0.66**	**5.14 +/- 2.37**	**14.53**	**FAO-56 Ground-Truth Reference Evapotranspiration**

3. Phase 2: Ground-Truth ET0 Formulation via FAO-56 Penman-Monteith

Machine learning models require a ground-truth supervision target (y). Since raw weather observations do not include ET0, we implemented the full physical FAO-56 Penman-Monteith equation (Allen et al., 1998):

```
ET0 = [ 0.408 * delta * (Rn - G) + gamma * (900 / (T + 273)) * u2 * (es - ea) ] / [ delta + gamma * (1 + 0.34 * u2) ]
```

Physical Breakdown of Sub-Components Calculated in Code:

1. **Psychrometric Constant (gamma):** Calculated as gamma = 0.000665 * P where P is measured atmospheric surface pressure in kPa (~98.3 kPa for Hisar at 215m altitude).
2. **Slope of Saturation Vapor Pressure Curve (delta):** Quantifies the rate at which air moisture capacity expands per degree rise: delta = 4098 * [0.6108 * exp(17.27 * T / (T + 237.3))] / (T + 237.3)^2.
3. **Vapor Pressure Deficit (es - ea):** Quantifies the drying power of the atmosphere. Saturation vapor pressures (e0_tmax, e0_tmin) are computed via Tetens equation, averaged to es, and actual vapor pressure is ea = es * (RH / 100).
4. **Extraterrestrial Solar Geometry (Ra):** Solar declination (delta_s), earth-sun distance factor (dr), and sunset hour angle (ws) are computed from the day of year (1-365) and station latitude (29.15 N).
5. **Net Radiation (Rn = Rns - Rnl):** Net shortwave solar radiation absorbed by green reference grass (albedo alpha = 0.23) is Rns = (1 - 0.23) * Rs. Net outgoing longwave heat loss (Rnl) is computed via the Stefan-Boltzmann law adjusted for humidity and cloud cover (Rs / Rso).
6. **Soil Heat Flux (G):** For daily time steps, G is standardly assumed to be 0 (G ~ 0).

Scientific Benchmarking & Verification:

We verified our pure NumPy calculation against the certified scientific hydrology library pyet (Python Evapotranspiration). Across all 5,113 days, the maximum discrepancy was 0.01 mm/day and the **mean absolute difference was 0.00012 mm/day**, confirming zero mathematical error in our target formulation.

4. Phase 2: Kaggle Notebook Architecture & Exploratory Insights

The entire implementation was consolidated into a modular, self-contained Jupyter notebook: evapotranspiration_fao56_calculation.ipynb. It contains 7 sections engineered for 100% reproducibility on Kaggle:

- **Universal Data Ingestion:** Automatically checks /kaggle/input/, local subdirectories, and downloads folders without path hardcoding.
- **Zero-Dependency Vectorization:** Avoids requiring internet access on Kaggle to pip-install external hydrology packages.
- **Automated Export & FileLink:** Exports the final clean dataset and uses IPython.display.FileLink to render a direct 1-click browser download link inside the notebook output.

Key Scientific Insights from Exploratory Visualizations:

1. **Multi-Year Seasonality (Time-Series Plot):** Daily ET0 displays an annual cyclic wave, dropping to 0.66 - 1.5 mm/day in December/January due to low solar angles and cold air, and spiking to 12 - 14.5 mm/day in May/June during intense pre-monsoon heatwaves.
2. **Seasonal Variance (Monthly Boxplot):** May and June exhibit the highest variance and peak medians (~10 mm/day). In July and August, despite high temperatures, ET0 drops to ~6 mm/day because monsoon clouds reduce incoming solar radiation and high humidity reduces the vapor deficit.
3. **Feature Correlation Ranking (Heatmap):** Solar radiation has the strongest positive correlation with ET0 (r = +0.81), followed closely by Temperature (r = +0.74). Relative Humidity exhibits a strong negative correlation (r = -0.58). This demonstrates that Temperature and Humidity together account for the dominant physics of evapotranspiration.

5. Roadmap: Transition to Phase 3 (Machine Learning Modeling)

With the data verified and ground-truth ET0 generated, the project is ready for **Phase 3 model training** across the 3 reduced-input configurations specified in the project document:

Configuration	Input Features (X)	Simulated Sensor Hardware	Cost Equivalent
Config 1: Temp Only	max_temp, min_temp	Dual thermistor / digital temp probe	< $5 USD (Ultra low-cost)
Config 2: Temp + Humidity	max_temp, min_temp, relative_humidity	Combined DHT22 / SHT31 sensor	< $10 USD (Low-cost standard)
Config 3: Temp + RH + Wind	max_temp, min_temp, RH, wind_speed	Temp/RH sensor + 3-cup anemometer	$50 - $150 USD (Moderate cost)
Full Benchmark (Upper Bound)	All inputs + Solar Radiation	Full weather station + precision pyranometer	> $1,500 USD (High cost)

Phase 3 Execution Steps: We will train **Linear Regression (baseline)**, **Random Forest Regressor (main model)**, and **Support Vector Regression (SVR)** across these configurations. Models will be evaluated on unseen test data using **MAE**, **RMSE**, and **R2**, followed by Random Forest Gini feature importance analysis to answer the core research question.